A RELATIVE H1-REGULAR IMMERSION OVER A LOCALLY FINITELY PRESENTED BASE IS FLAT ON A NEIGHBOURHOOD OF ITS IMAGE AND IS A REGULAR IMMERSION STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 5395-5406 and proof span 5408-5484 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof names its approximating rings with the author's own indexing and states the colimit of the Koszul complexes; both are preserved.

Editorial note. The proof contains the sentence 'Assume f_i is the image of f_i' for the lifts of the sequence; the author's phrasing is kept unchanged.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The approximation of a flat and finitely presented algebra by a flat algebra of finite type over a finitely generated ring, cited from the algebra chapter. Cited at proof lines 5426-5429. Produces the Noetherian model over which the quotient by the chosen sequence is flat.
- Surjectivity of the base change map on the first Koszul cohomology of a sequence, and exactness of directed colimits, both cited from the algebra and more-algebra chapters. Cited at proof lines 5437-5439 and 5450-5452. Lets the finitely many generators of the Koszul cohomology of the model be killed after a further finitely generated extension of the base ring.
- Flatness descends along the base change defining the model, cited from the morphisms chapter. Cited at proof lines 5462-5464. Transfers the flatness statement obtained for the model back to the original morphism.
- A quasi-regular immersion is flat at each point of its image, and the flat locus of a finite type morphism of Noetherian schemes is open, cited from the same chapter and the algebra chapter. Cited at proof lines 5468-5473.

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Supplies pointwise flatness in the reduced situation and turns it into an open neighbourhood.
- A relative regular immersion which is flat in a neighbourhood of its image is a regular immersion with the property stable under base change, cited from the same chapter. Cited at proof lines 5474-5476. Gives part (2) once part (1) is known.

Source reference. Original lemma 00DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 491–491. Same-chapter reference used in the retained author text at source line(s) 5458. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3343–3343. Same-chapter reference used in the retained author text at source line(s) 4106, 4234, 5430. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original theorem 00RC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 998–998. Same-chapter reference used in the retained author text at source line(s) 4132, 4252, 5481. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7842–7842. Same-chapter reference used in the retained author text at source line(s) 5479. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7843–7843. Same-chapter reference used in the retained author text at source line(s) 5301, 5387, 5483, 5511. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7839–7839. Same-chapter reference used in the retained author text at source line(s) 4113, 5441. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01U8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2401–2401. Same-chapter reference used in the retained author text at source line(s) 4126, 5474. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes. Let $Z \to X$ be a relative H_1-regular immersion. Assume $X \to S$ is locally of finite presentation. Then*

1. *there exists an open subscheme $U \subset X$ such that $Z \subset U$ and such that $U \to S$ is flat, and*
2. *$Z \to X$ is a regular immersion and the same remains true after any base change.*

Proof. Pick $x \in Z$. To prove (1) suffices to find an open neighbourhood $U \subset X$ of x such that $U \to S$ is flat. Hence the lemma reduces to the case that $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine and that Z is given by an H_1-regular sequence $f_1, \ldots, f_r \in B$. By assumption B is a finitely presented A-algebra and $B/(f_1, \ldots, f_r)B$ is a flat A-algebra. We are going to use absolute Noetherian approximation.

Write $B = A[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$. Assume f_i is the image of $f_i' \in A[x_1, \ldots, x_n]$. Choose a finite type $\mathbf{Z}$-subalgebra $A_0 \subset A$ such that all the coefficients of the polynomials $f_1', \ldots, f_r', g_1, \ldots, g_m$ are in A_0. We set $B_0 = A_0[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$ and we denote $f_{i,0}$ the image of f_i' in B_0. Then $B = B_0 \otimes_{A_0} A$ and

$$B/(f_1, \ldots, f_r) = B_0/(f_{0,1}, \ldots, f_{0,r}) \otimes_{A_0} A.$$

By Algebra, Lemma 02JO we may, after enlarging A_0, assume that $B_0/(f_{0,1}, \ldots, f_{0,r})$ is flat over A_0. It may not be the case at this point that the Koszul cohomology group $H_1(K_\bullet(B_0, f_{0,1}, \ldots, f_{0,r}))$ is zero. On the other hand, as B_0 is Noetherian, it is a finitely generated B_0-module. Let $\xi_1, \ldots, \xi_n \in H_1(K_\bullet(B_0, f_{0,1}, \ldots, f_{0,r}))$ be generators. Let $A_0 \subset A_1 \subset A$ be a larger finite type $\mathbf{Z}$-subalgebra of A. Denote $f_{1,i}$ the image of $f_{0,i}$ in $B_1 = B_0 \otimes_{A_0} A_1$. By More on Algebra, Lemma 063Q the map

$$H_1(K_\bullet(B_0, f_{0,1}, \ldots, f_{0,r})) \otimes_{A_0} A_1 \longrightarrow H_1(K_\bullet(B_1, f_{1,1}, \ldots, f_{1,r}))$$

is surjective. Furthermore, it is clear that the colimit (over all choices of A_1 as above) of the complexes $K_\bullet(B_1, f_{1,1}, \ldots, f_{1,r})$ is the complex $K_\bullet(B, f_1, \ldots, f_r)$ which is acyclic in degree 1. Hence

$$\mathrm{colim}_{A_0 \subset A_1 \subset A} H_1(K_\bullet(B_1, f_{1,1}, \ldots, f_{1,r})) = 0$$

by Algebra, Lemma 00DB. Thus we can find a choice of A_1 such that $\xi_1, \ldots, \xi_n$ all map to zero in $H_1(K_\bullet(B_1, f_{1,1}, \ldots, f_{1,r}))$. In other words, the Koszul cohomology group $H_1(K_\bullet(B_1, f_{1,1}, \ldots, f_{1,r}))$ is zero.

Consider the morphism of affine schemes $X_1 \to S_1$ equal to Spec of the ring map $A_1 \to B_1$ and $Z_1 = \mathrm{Spec}(B_1/(f_{1,1}, \ldots, f_{1,r}))$. Since $B = B_1 \otimes_{A_1} A$, i.e., $X = X_1 \times_{S_1} S$, and similarly $Z = Z_1 \times_S S_1$, it now suffices to prove (1) for $X_1 \to S_1$ and the relative H_1-regular immersion $Z_1 \to X_1$, see Morphisms, Lemma 01U8. Hence we have reduced to the case where $X \to S$ is a finite type morphism of Noetherian schemes. In this case we know that $X \to S$ is flat at every point of Z by Lemma 063T. Combined with the fact that the flat locus is open in this case, see Algebra, Theorem 00RC we see that (1) holds. Part (2) then follows from an application of Lemma 063U. □